

Effect of training programs on the creativity of university students: A multi-level meta-analysis

Huifen Guo^a, Zhen Zhou^{b,*}, Fengqi Ma^a

^a School of Education, Guangzhou University, Guangzhou, Guangdong, China

^b School of Mechanical and Electrical Engineering, Guangzhou University, Guangzhou, Guangdong, China

ARTICLE INFO

Keywords:

Creativity
Meta-analysis
Training program
University students

ABSTRACT

In today's fast-paced and technology-driven world, creativity is crucial for individuals and institutions. However, studies on the methods employed for cultivating creativity through training programs and the magnitude of their impact are limited. This study conducted a meta-analysis to comprehensively elucidate the influence of training initiatives on enhancing higher education students' creativity across various domains. Additionally, the effects of pivotal moderators within these training programs were examined. A comprehensive meta-analysis, encompassing 29 empirical studies published between 2000 and 2024 and 109 effect sizes, and moderation analyses focused on various key variables were conducted. The findings indicated that creativity training programs could effectively cultivate diverse facets of students' creativity, including creative thinking, graphic creativity, verbal creativity, word creativity, emotional creativity, and problem-solving. Overall, the results revealed a medium yet statistically significant impact of training programs on the creativity of university students (Hedges' $g = 0.628$, $p = 0.000 < 0.01$). These findings offer valuable insights for educators and policymakers in formulating curricula and interventions aimed at fostering students' creativity.

1. Introduction

A highly creative society represents untapped potential and unprecedented possibilities. As the driving force behind national progress, university students play a pivotal role in shaping the future, and governments globally are prioritizing the cultivation of their creativity (Liu et al., 2023). Given the fundamental importance of creativity within higher education, it is imperative to elucidate the cognitive and pedagogical mechanisms that underpin creativity training for university students. Such insights are essential for the development of evidence-based training interventions designed to optimize their creative performance (Yeh & Lin, 2015). Previously limited to certain disciplines such as visual arts or music, creativity is now regarded as inherently intertwined with all aspects of cognition (Khalil et al., 2019). Consequently, in recent decades, there has been growing interest in the study of creativity across various research fields (Ruiz-del-Pino et al., 2022; Sio & Lortie-Forgues, 2024). With the growing recognition of the importance of creativity across an array of sectors, especially with innovation, there is a rising trend toward enhancing creativity through training programs (Onarheim & Friis-Olivarius, 2013; Scott et al., 2004).

Is creativity trainable? Torrance (1972) synthesized findings from 142 studies aimed at assessing methods for fostering creative

* Corresponding author at: 230 Waihuan West Road, Higher Education Megacenter, Panyu District, Guangzhou City, Guangdong Province, 510006, China.

E-mail address: zhouzhen@gzhu.edu.cn (Z. Zhou).

<https://doi.org/10.1016/j.tsc.2025.101779>

Received 25 April 2024; Received in revised form 14 December 2024; Accepted 25 January 2025

Available online 27 January 2025

1871-1871/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

thinking in children and determined that the Osborn–Parnes creative problem-solving (Osborn, 1963; Parnes, 1967) training program and its adaptations yielded the most favourable outcomes. Mansfield et al. (1978) concluded that “most assessments of creativity training initiatives seem to validate the notion that creativity is trainable”. However, they expressed reservations regarding the transferability of training effects to different, untrained tasks or to real-world creative endeavours. Gu et al. (2024) conducted a study using a pre/posttest between-subjects design with two control groups: one undergoing unrelated online memory training and the other receiving no training. Four established tasks assessing divergent and convergent thinking were used to measure participants’ creativity. After completing the online creativity training, participants showed significant improvements in various aspects of divergent thinking (e.g., fluency, flexibility, and originality). Cheung and Hui (2023) examined the effectiveness of an art-centred training program on students’ creative disposition. Compared with the control group, the experimental group showed positive changes in creativity between pre- and post-tests after undergoing six training program sessions.

As mentioned, there is a well-established understanding that creativity training is effective. However, determining the most effective methods remains challenging owing to conflicting findings in previous comprehensive reviews (Chen et al., 2014; Im et al., 2015; Ritter & Mostert, 2017; Valgeirsdottir & Onarheim, 2017). This study sought to identify the factors that differentiate the effectiveness of creativity training programs. This will provide a more robust estimate of the average effect size of creativity training, offering researchers critical insights into existing methodological limitations within the field. This enhanced understanding will inform the development of empirically grounded training programs, which rely on rigorous, evidence-based approaches. Additionally, our findings will support educational leaders in making data-driven decisions regarding creativity enhancement strategies, enabling them to set more realistic expectations for the outcomes of such initiatives. On a theoretical level, the empirical findings of this research validate the theoretical underpinnings for focusing on cognitive interventions to enhance creativity. The study also provides a comprehensive framework for designing educational strategies and curricula that are geared towards developing creative abilities. On a practical level, the research offers tangible insights for educators and curriculum developers to implement in their teaching practices. Moreover, it outlines specific strategies and tools that can be seamlessly integrated into current educational frameworks, thus promoting the enhancement of students’ creativity.

2. Literature review

2.1. Concept of creativity

The concept of creativity is multifaceted, with scholars from various fields emphasizing different aspects of its study. Some researchers focus on creative personality traits, others on the creative process, while still others examine creative outcomes. To gain a more comprehensive understanding of creativity, this study categorizes its definition into four distinct types, drawing on existing literature.

The first category of scholars conceptualizes creativity as a constellation of individual personality traits. For instance, Amabile (1983) highlights that individuals who exhibit high levels of creativity often possess traits such as independent thinking, self-efficacy, an affinity for complexity, aesthetic sensitivity, and a propensity for risk-taking. Sternberg (1986) contends that creative individuals are distinguished by their need for social validation, cognitive flexibility, tolerance for ambiguity, resilience in the face of adversity, and a moderate inclination towards risk-taking behavior. Davis (2004) proposes that creativity is a composite of interrelated individual characteristics, including openness to experience, diverse interests, a willingness to challenge established norms, and a high degree of autonomy.

The second category of scholars conceptualizes creativity in terms of creative outcomes. Bruner (1962) posits that creativity is evidenced by products that not only receive acknowledgement from observers for their novelty and appropriateness but also evoke a sense of “genuine astonishment.” Plucker et al. (2004) suggest that creativity is the result of the dynamic interplay between cognitive abilities, creative processes, and environmental factors, culminating in a tangible product that is both original and functional, as delineated within the socio-cultural context. Karwowski and Soszynski (2008) define creativity as the generation of unexpectedly innovative outputs that effectively achieve a specific functional objective.

The third category of scholars posits that creativity is fundamentally rooted in the creative process. Wallas (1926) delineates creativity as unfolding through a sequence of stages in problem-solving, specifically preparation, incubation, illumination, and verification. Puccio et al. (2010) further refines this view by characterizing creativity as a process of problem resolution, which involves stages such as problem identification, idea generation, idea evaluation, and solution validation. Cropley (2016) advances a model comprising seven interrelated stages: preparation, activation, generation, illumination, verification, communication, and validation. This model underscores the dynamic and iterative nature of creativity, reflecting its complex evolution through various phases of cognitive and evaluative activities.

The fourth category of scholars examines creativity through cognitive methodologies (Grégoire et al., 2011). Freud (1910, 1964) theorizes that creativity arises from the dynamic tension between conscious reality and unconscious drives. Finke et al. (1992) assert that creativity is a modest extension of research into perceptual and mnemonic systems. Weisberg (1993) contends that creativity is essentially a typical cognitive process that yields remarkable products. Findlay and Lumsden (1988) characterize creativity as a cognitive and associative capability.

We have synthesized the concept of creativity, considering it a complex construct that encompasses intricate psychological processes (Hennessey & Amabile, 2010), including creative thinking, personality, motivation, cognition, and more (Lucchiari et al., 2019; Said-Metwaly et al., 2021; Van der Zanden et al., 2020). Evidence suggests that enhancements in creativity can be achieved through training (Gutman & Schoon, 2013).

2.2. Creativity training programs

Given the importance of creativity, efforts have been made to identify efficient methods for enhancing it. Many approaches to enhancing creativity have been developed, and both organizations and educational institutions have invested time and resources in developing and implementing creativity training. The first recorded attempt to develop creative skills in individuals was in 1937, with the pioneering of the Creative Engineering Program by A.R. Stevenson, which was the world's first formal course dedicated to creativity training (Samstad, 1962; Von Fange, 1963). Since numerous studies have been conducted on creativity training, GE's consistent introduction of programs like the Leadership Innovation and Growth (LIG) initiative, which was launched in 2006, serves as evidence of the ongoing research in this field (Xu, 2013). Furthermore, several meta-analyses have compared the findings of these individual studies on training efforts (Scott et al., 2004). Numerous scholars have formulated definitions for creativity training programs. Rose and Lin (1984) state that "all these approaches share a common assumption that training, practice, and encouragement in utilizing creative thinking skills can enhance individuals' creativity". Scholars contend that creativity can be conceptualized as a term commonly used to describe a range of initiatives aimed at enhancing creativity (Onarheim & Friis-Olivarius, 2013; O'Connor et al., 2016). A creativity training program is a predefined and structured regimen comprising one or multiple sessions, with the primary aim of amplifying the creativity of one or multiple participants (Valgeirsdottir & Onarheim, 2017). This definition highlights the importance of programs being predefined and structured with scheduled sessions to enhance participants' creativity.

Over the past few years, educators and psychologists have developed various methodologies and instructional resources to foster creativity. They classified them based on the characteristics of the training program. Mansfield et al. (1978) qualitatively assessed the effectiveness of creativity training programs and categorized them into six groups: the productive thinking program, which emphasizes both divergent and convergent thinking skills; the Purdue creative thinking program, accompanying written exercises for divergent thinking skill practice; the partner program, such as brainstorming; the Myers–Torrance workbooks, which offer exercises to enhance perceptual and cognitive abilities; Khatena's training method, involving the teaching and practice of five thinking skills (i.e., deviation, transposition, analogy, restructuring, and synthesis); and other training programs such as synectics, using various thinking skills to facilitate. Fleith et al. (2002) stated that major creativity training programs include the creative problem-solving program, the productive thinking program, the Purdue creative thinking program, and new directions in creativity. According to Dow and Mayer (2004), training programs can be classified into various types, including verbal training, cognitive training, technique-based training, task-based training, emotional training, mathematical training, spatial training, and combined training. Ma (2006) argued that types of creativity training programs could be classified as follows: simple ideation training, brainstorming, incubation, forced relation, part improving, morphological synthesis, attitude training, synectics, and idea checklist. Valgeirsdottir and Onarheim (2017) categorized the training program into the following types: embodied creativity training, the creativity course, short-term meditation, creative problem-solving, creative drama process, creativity exercises, role-play training, creativity capacity-building program, creativity development and application, creative self-efficacy, applied neuro creativity, creative thinking program, innovation boot camp, creativity scripts, and creativity training.

In this study, we referred to the categorization of creativity training proposed by Dow and Mayer (2004). This categorization, grounded in empirical cluster analysis, facilitates a precise evaluation of the effectiveness of targeted interventions. Research by Mayer (2006) and Kolayis et al. (2012), illustrates this point. The first author categorizes the projects according to the designated characteristics, followed by a review by the co-author to ensure the consistency of the classification results.

Considering prior deliberations, the objective of this work was to identify, delineate, assess, and amalgamate research on the effectiveness of training programs for enhancing creativity among university students. Therefore, the research questions to be addressed were as follows:

RQ1: What are the key attributes of effective creativity training?

RQ2: Which groups show the most significant improvements in creativity following training?

RQ3: What forms of creativity training are most effective according to empirical evidence?

3. Research method and process

3.1. Data sources and search strategies

To ensure the rigor and clarity of our literature extraction process, we conducted a systematic review following the guidelines of the preferred reporting items for systematic reviews and meta-analyses (PRISMA) procedure (Moher et al., 2015). Considering that the objective of the PRISMA declaration is to assist writers in improving the presentation of systematic reviews and meta-analyses, we followed these criteria for our systematic review.

The review team aimed to assemble a comprehensive body of studies offering an objective evaluation of creativity training programs for university students. The literature search was conducted in two phases. The first phase employed systematic Boolean searches across multiple electronic platforms and databases, using diverse search terms. This initial search was completed in January 2024. The chosen timeframe for our investigation was the 2000–2024 period. The second phase involved supplementary techniques such as manual website searches, literature snowballing, and consultations with field experts.

Following Moher et al. (2015), at least two scientifically credible databases were consulted during the search and selection phases. These databases, known for their academic rigor and broad disciplinary coverage, ensured an extensive and unbiased search scope (Zhang et al., 2024). The decision to apply the open-access filter was based on the accessibility criterion by Moreau and Gamble (2022), emphasizing the need for transparency and replicability, as supported by López-Nicolás et al. (2022). This approach also aligns with

the Comprehensive Search guidelines of [Pigott and Polanin \(2020\)](#).

Search keywords were selected following the recommendations of [Ruiz-del-Pino et al. \(2022\)](#), balancing sensitivity and specificity. The search strategies were adapted to each platform's functionality. For databases with advanced search capabilities, terms were classified into independent variables, dependent variables, study design, and participant population, then combined using "AND" and excluded with "NOT" to refine results. For databases with limited search features, the team adjusted the strategy, often using combinations of keyword and topic searches ([Kugley et al., 2016](#)).

Fig. 1. illustrates the process of article selection, depicting the search procedure. Following the application of eligibility criteria and the scrutiny of the records, 29 studies were included.

3.2. Literature inclusion and exclusion criteria

To ensure the accuracy and validity of the research, we established the following criteria for study selection. (a) Theme: We focused on studies that addressed the enhancement of university students' creativity through training programs, without differentiating between domain-specific creativity development (e.g., creative writing) and general cognitive creativity improvement (e.g., divergent thinking skills). (b) Category: We only considered experimental research, including randomized and quasi-experimental studies. For single-group trials, a clear pre/posttest was required, while dual-group trials needed to include both experimental and control groups. (c) Setting: We considered studies in formal classroom instruction and informal extracurricular instruction settings to ensure diverse research contexts. (d) Participants: Our research subjects were limited to university students undergoing a training program. (e) Results: Our primary focus was on various domains of creativity, such as verbal creativity, emotional creativity, and graphic creativity.

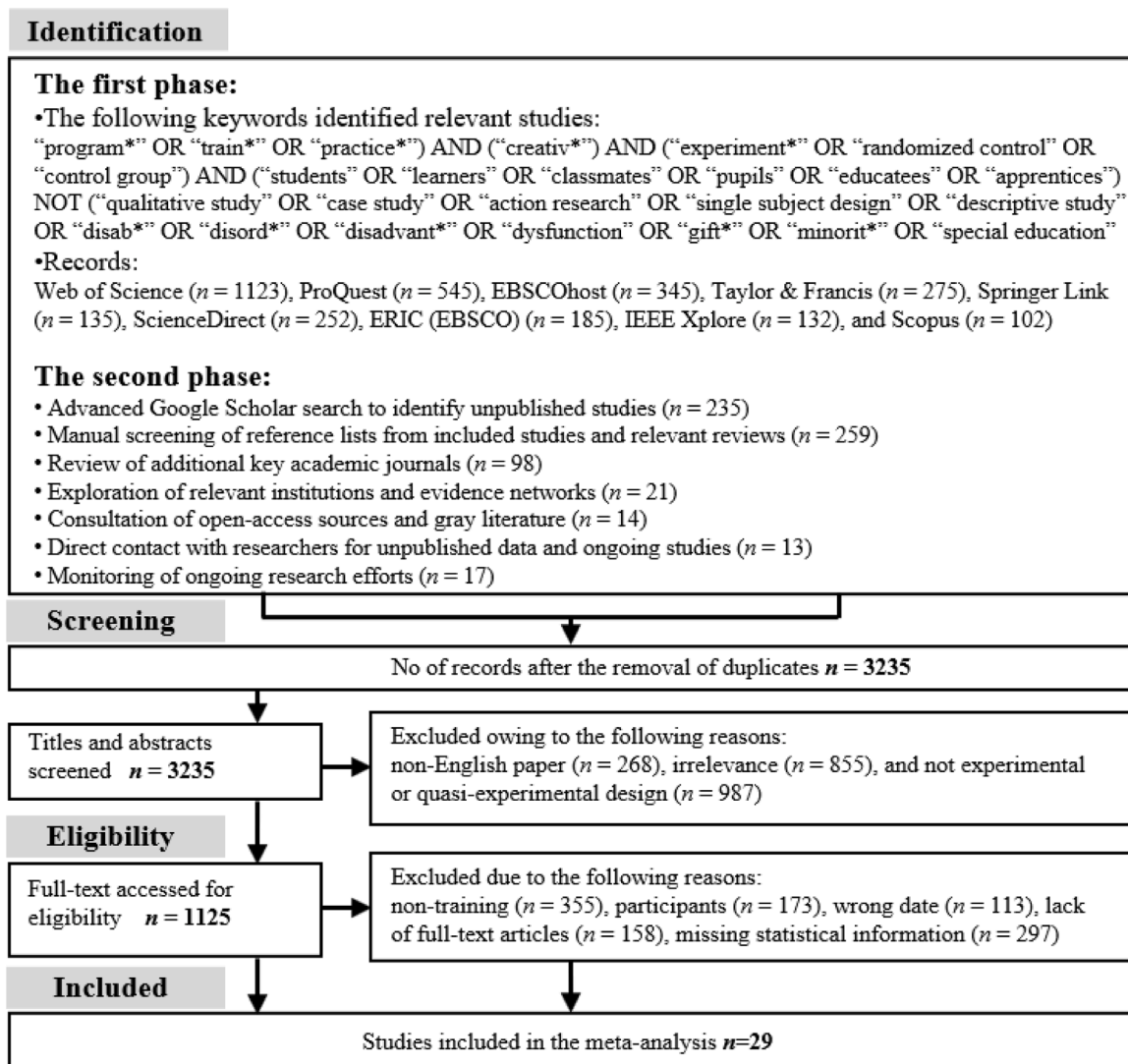


Fig. 1. PRISMA flowchart of included studies.

To enhance the persuasiveness and practical value of our meta-analysis results, we employed the following exclusion criteria to select the most relevant and highest-quality studies for our research topic. (a) Data: We only considered studies with clear and comprehensive data, such as means and standard deviations, which are essential for effective meta-analysis. (b) Relevance: We excluded studies that did not focus on describing creativity training programs, such as studies on enhancing creativity through manipulation, training planning methods, and diverse interventions. (c) Filtering: We excluded studies that were duplicates or secondary publications of the same data. (d) Accessibility: We excluded studies that were unavailable in full text or that were in a language that we could not understand.

3.3. Literature coding

Data and information from a selected subset of studies were independently extracted and coded by two evaluators, who resolved any discrepancies, and validation was conducted by another member of the research team. Following the PRISMA recommendations for systematic review protocols (Moher et al., 2015), the extracted data included author names, country, type of measure, type of administration, sample size, activity for the control group, duration, activity type, and training procedure. Subsequently, a comprehensive coding list was devised according to the extracted information, as depicted in Appendix A (Table 4). All studies selected for inclusion were carefully read and coded.

3.4. Statistical analysis

We initially input the collected data into the Comprehensive Meta-Analysis (CMA) Version 3.070 software, which can be accessed at <https://meta-analysis.com/>. The procedure is as follows: click on “Insert” and select “column for...” where you will choose “Study name”, “Subgroups within study”, “Comparison names”, and “Outcome names”. Subsequently, based on statistical requirements, set the Column function for variables in the “Column format”, such as “Moderator”, “Time point”, and so forth. Once the parameters for the variables have been set, clicking on “Run analyzes” will yield the analytical results of the data. We reviewed and revised these results. Once all necessary adjustments were made (i.e., regarding data formats and analysis methods), the output was recorded.

Effect size, a common measure of the strength and direction of the relationship between two variables or the level of practical impact, is often calculated using Cohen's d and Hedges' g in meta-analysis (Borenstein et al., 2010). Hedges' g and Cohen's d are highly comparable measures of effect size, with Hedges' g often referred to as the corrected effect size. For very small sample sizes, typically fewer than 20, Hedges' g is preferred due to its adjustment for bias. When the sample size exceeds 20, the results generated by both statistics tend to be approximately equivalent, yielding similar estimates of effect size (Borenstein et al., 2009; Card, 2012). This study includes 29 samples, thus Cohen's conventional interpretation of effect size is employed: an effect size of 0.2 is considered small, 0.5 is regarded as medium, and 0.8 is classified as large (Cohen, 1977).

The statistical implications of these effect sizes are typically presented with a 95 % confidence interval. The farther from zero, the stronger the impact. A zero mean effect indicates the absence of a significant relationship between the studied variables. To account for the diversity of research methodologies, group sizes, training techniques, and comparison groups among the selected studies, a random-effects model was used (Cohen, 1977).

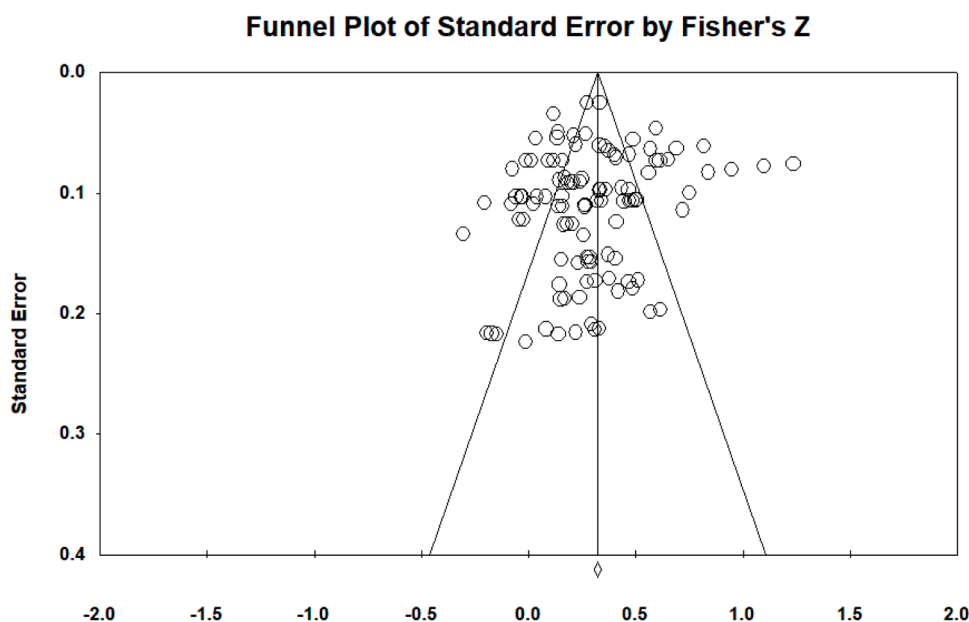


Fig. 2. Funnel plot of standard error by Fisher's Z.

We used the I^2 test to uncover variations in effect sizes stemming from discrepancies among studies (Borenstein et al., 2021). An I^2 value of 0 % indicates a lack of heterogeneity, whereas an I^2 value of <25 % indicates slight heterogeneity. An I^2 value exceeding 25 % but below 50 % indicates a moderate degree of heterogeneity, and an I^2 above 50 % indicates substantial heterogeneity (Higgins et al., 2003).

To assess the potential impact of publication bias on the findings, the initial step involved constructing a funnel plot based on Hedges' g (random effects; Egger et al., 1997). Additionally, determining the existence of symmetry involved calculating Begg and Mazumdar correlations. To estimate the fail-safe N using Rosenberg's method, the value X was computed, which should exceed 5k (the count of independent effect sizes) +10 (Rosenthal, 1979).

4. Research results

4.1. Publication bias analysis

It is essential to address publication bias before conducting a meta-analysis. Publication bias, a form of bias, occurs when experimental or research outcomes influence the decision to publish. It describes the practice in which "research with favourable outcomes tends to have a higher chance of publication and is often published more quickly than studies with unfavourable results" (Dwan et al., 2008; Murad et al., 2018). To evaluate if publication bias could undermine the validity of the meta-analysis results, a funnel plot was created. Fig. 2 presents the funnel plot illustrating the 109 effect sizes. The majority of the distribution clusters are near the top of the plot and close to the mean. However, the lower portion of the funnel plot does not show clear asymmetry on its right side. Additionally, some studies deviate from the average effect sizes. Consequently, the likelihood of publication bias in this study was minimal.

4.2. Estimation of effect sizes and heterogeneity analysis

In this meta-analysis, 109 effect sizes were obtained from the analysis of 29 separate interventions. As shown in Table 1, the random model indicated a statistically significant and moderately positive impact of training programs on the creativity of university students. Examining heterogeneity is crucial to amalgamate the global effect value from the effectiveness of individual studies. Heterogeneity in meta-analysis is defined as "the variability in study outcomes among studies" (Borenstein et al., 2009). In this study, heterogeneity was assessed using the Q test. In general, in meta-analysis, the Q and I^2 values serve as criteria for assessing the level of heterogeneity among effect values. If Q surpasses $df(Q)$ and I^2 exceeds 75 %, both with significance levels below 0.05, the random effect model is typically employed; otherwise, the fixed-effects model is used. The results for each relationship scrutinized in the meta-analysis are depicted in Table 1. This study rejected the null hypothesis of homogeneity ($I^2 = 84.31 \% > 75 \%$). Although the overall effect size (0.628) indicated a moderate correlation, it remained significant ($p > 0.1$) for practical purposes (Kansizoglu, 2017).

Table 2 illustrates the impact of training programs on various subdomain creativity measures. The overall effect size was Hedges' $g = 0.569$ (95 % $CI = 0.480-0.658$), indicating a statistically significant medium effect size. Among these effect sizes, "creative thinking" displayed the most significant effect (Hedges' $g = 0.671$, $p < 0.01$). The subdomains that received relatively more emphasis were "graphic creativity" and "word creativity" (Hedges' $g = 0.644$, $p < 0.01$; Hedges' $g = 0.641$, $p < 0.01$, respectively). "Verbal creativity" and "problem-solving" attracted attention and exhibited a moderate effect (Hedges' $g = 0.606$, $p = 0.00 < 0.01$; Hedges' $g = 0.539$, $p < 0.01$, respectively). "Emotional creativity" demonstrated a low effect (Hedges' $g = 0.397$, $p < 0.01$).

4.3. Moderator analysis

In CMA, we clicked the "group by" button and tested the moderating effects of "duration," "academic level," "activity type," and "training type" as shown in Table 3. The Hedges' g for "< 2 months" was the highest. Although the Hedges' g for "< 4 months" was higher than those for "< 2 years" and "< 1 week," they all exhibited a medium moderating effect, with "< 3 years" showing the weakest moderation. According to our analysis of "duration," we identified a curvilinear relationship between the duration of training and its efficacy, characterized by an inverted U-shaped relationship. Initially, the training effectiveness increased with time. However, beyond a critical threshold, prolonged training duration led to a decline in effectiveness.

Our data analysis indicated that Hedges' g for "graduate" (0.991) exceeded that for "undergraduate" (0.613). To better understand

Table 1
Random model for overall effects size.

Model	Number of Studies	Effect size and 95 % confidence interval				Test of null (two-tail)			
		PE	SE	V	LL	UL	Z	P	
Fixed	29	0.572	0.034	.001	0.504	0.639	16.645	.000	
Random	29	0.628	0.092	.009	0.446	0.809	6.788	.000	
Model	Number of Studies	Heterogeneity	τ^2						
		Q	$df(Q)$	P	I^2	τ^2	SE	V	τ
Fixed	29	178.507	28	0.000	84.314	0.189	0.075	0.006	0.435
Random	29								

Note: CI = confidence interval; PE = point estimate; V = variance; LL = lower limit; UL = upper limit.

Table 2

The effect size of different outcomes.

Outcomes	Effect size and 95 % CI				Test of null			
	N	Hedges' g	SE	V	LL	UL	Z	P
Creative thinking	24	0.671	0.118	0.014	0.439	0.903	5.675	0.000
Graphic creativity	27	0.644	0.105	0.011	0.439	0.849	6.153	0.000
Verbal creativity	23	0.606	0.102	0.010	0.407	0.805	5.966	0.000
Words creativity	16	0.641	0.154	0.024	0.339	0.943	4.156	0.000
Emotional creativity	9	0.397	0.097	0.009	0.208	0.587	4.106	0.000
Problem-solving	10	0.539	0.114	0.013	0.316	0.762	4.730	0.000
Total effect size	109	0.569	0.045	0.002	0.480	0.658	12.544	0.000

Note: CI = confidence interval; SE = standard error; V = variance; LL = lower limit; UL = upper limit.

Table 3

The effect size of moderator variables.

Categories	Effect size and 95 % CI				Test of null				
	N	Hedges' g	SE	V	LL	UL	Z	P	
Duration	<1week	37	0.576	0.078	0.006	0.424	0.728	7.415	0.000
	<2months	30	0.844	0.101	0.010	0.647	1.042	8.382	0.000
	<4months	24	0.642	0.101	0.010	0.444	0.840	6.356	0.000
	<1year	9	0.233	0.076	0.006	0.085	0.381	3.083	0.002
	<2years	4	0.418	0.092	0.009	0.237	0.599	4.518	0.000
	<3years	5	-0.122	0.190	0.036	-0.493	0.250	-0.641	0.522
Academic level	Undergraduate	107	0.613	0.049	0.002	0.517	0.709	12.502	0.000
	Graduate	2	0.991	0.267	0.071	0.468	1.515	3.712	0.000
Activity type	Group	61	0.567	0.054	0.003	0.462	0.673	10.539	0.000
	Alone	48	0.682	0.092	0.009	0.501	0.863	7.375	0.000
Training type	Verbal training	15	0.998	0.286	0.082	0.437	1.560	3.485	0.000
	Thinking training	13	0.762	0.149	0.022	0.469	1.055	5.102	0.000
	Technique-based training	20	0.552	0.121	0.015	0.314	0.789	4.550	0.000
	Task-based training	5	0.685	0.195	0.038	0.302	1.068	3.503	0.000
	Emotional training	2	-0.066	0.171	0.029	-0.401	0.269	-0.384	0.701
	Combined training	54	0.513	0.053	0.003	0.410	0.617	9.728	0.000

Note: V = variance; LL = lower limit; UL = upper limit.

the differences in training outcomes between undergraduate and graduate students, we conducted an analysis comparing the efficacy of these two cohorts. The findings indicated that graduate students exhibited higher training outcomes than their undergraduate counterparts. However, the overall mean effect size was heavily influenced by the number of studies included in each subgroup. Therefore, our next step was to conduct additional literature searches focusing on studies involving graduate student participation in training programs, to incorporate additional data to analyze the disparities in effectiveness between the two groups.

Hedges' g for "alone" (0.682) exceeded that for "group" (0.567), indicating a significant impact of training format on creativity, with individual training yielding better results. From the perspective of training format, individual training was more effective than group training. The discrepancy between these two approaches was minimal, indicating that the effectiveness of training primarily depends on its content and the measurement methods used. Individual training of participants has the potential to enhance training effectiveness.

Regarding training methods, "verbal training" (0.998) showed the highest effectiveness, while "emotional training" (-0.066) showed the lowest effectiveness. Following "verbal training," "thinking training" (0.762) ranked next in effectiveness. Moreover, "task-based training" (0.685) was more effective than "technique-based training" (0.552) and "combined training" (0.513). This data presentation section highlights the success rates of different methods. The findings revealed that among the training modalities, "verbal training" was the most effective, presumably because language allows undergraduate students to directly perceive encouragement and support, thereby enhancing their confidence. "Thinking training" also demonstrated favorable outcomes, indicating that training students' thinking processes can stimulate them to generate more associations, thereby breaking through fixed thinking patterns and generating a variety of novel ideas. "Task-based training" exhibited comparable effectiveness to "technique-based training." This analysis suggests that external influences have a limited impact on students' creativity; the key lies in challenging students' fixed mindsets and altering their traditional viewpoints.

4.4. Meta-Regression analysis

We tested the effect of "duration" on creativity through meta-regression analysis. After the incorporation of "duration" into the model, τ decreased from 0.551 to 0.477, indicating a decrease of 0.074. This decrease represents a proportion of $0.074 / 0.551 = 0.134$ in the overall context, indicating that "duration" explained 13.4 % of the between-group variance. Therefore, the duration of training

exhibited a relatively minor impact on the enhancement of creativity. The results of the meta-regression analysis and moderation analysis were consistent, reinforcing that achieving better training outcomes depends more on content than duration.

5. Discussion and conclusion

5.1. Discussion

5.1.1. Regarding question 1: what are the key attributes of effective creativity training?

The analysis of “duration” suggests that opting for a duration aligned with training programs, rather than adopting an extended timeframe (3 years), yields the most gains in training effectiveness. Previous studies further support this conclusion (Liu et al., 2022; Park, 2023; Şendağ et al., 2023; and Xing & Qi, 2023). This phenomenon is attributable to several factors. First, there is the issue of forgetting and decay. Over time, individuals may gradually forget the content learned during the training. Lack of regular practice and reinforcement can lead to a decline in knowledge and skills (Ritter et al., 2020). Insufficient application and practice can contribute to this decline. Trainees need opportunities to apply their acquired skills and knowledge in real-world scenarios to maintain proficiency over time. Lack of sustained practice and application can diminish the effectiveness of creativity skills. Second, organizational culture and environment play a significant role. An organizational culture that does not support creative thinking or innovation can quickly diminish the effects of training (Chahar et al., 2019). Finally, prolonged interventions and training aimed at fostering students’ creativity have resulted in the development of certain latent qualities such as logical reasoning and problem-solving abilities. However, researchers have not reported changes in these latent qualities.

5.1.2. Regarding question 2: which groups show the most significant improvements in creativity following training?

The superior outcomes of creativity training among graduate students than among undergraduates are attributable to several key factors. First, graduate students typically possess higher cognitive maturity and critical thinking skills than undergraduate students (Dodds et al., 2010). Their advanced academic experiences equip them with a deeper understanding of complex concepts and a greater capacity for abstract thinking, which are essential components of creativity training. Second, graduate students often have more specialized knowledge within their respective fields of study (Nandi et al., 2000). This specialized knowledge provides a solid foundation upon which to build creative problem-solving skills and innovative thinking. They can leverage their in-depth understanding to effectively generate novel ideas and solutions during creativity training (Rampersad & Patel, 2014). Moreover, the graduate-level learning environment typically fosters a higher level of collaboration and intellectual discourse. Engaging with peers and faculty members in scholarly discussions and collaborative projects can stimulate creativity and inspire innovative thinking among graduate students (Matraeva et al., 2020).

The effectiveness of individual creativity training compared with group-based training can be elucidated through several pertinent factors. Individual creativity training offers a conducive environment for introspection and deep concentration. Participants can engage in tasks and exercises without external distractions, enabling them to fully immerse themselves in the creative process. This solitary focus facilitates a deeper exploration of personal creativity and encourages participants to freely generate ideas without the influence of group dynamics (Paulus, 2000). Furthermore, individual training allows for flexible pacing and scheduling, accommodating participants’ unique learning styles and preferences. Individuals can progress through the training program at their own pace, allowing them to fully grasp concepts and practice techniques at a comfortable rhythm. This flexibility promotes deeper engagement and retention of learning, maximizing training effectiveness.

5.1.3. Regarding question 3: what types of creativity training are most effective according to empirical evidence?

The variation in effectiveness among different types of creativity training—verbal, thinking, emotional, technique-based, task-based, and combined—is attributable to the unique focus of each approach on cognitive and psychological processes. *Verbal training* stands out as the most effective owing to its emphasis on enhancing verbal fluency, communication skills, and linguistic creativity. By encouraging participants to articulate their thoughts, ideas, and emotions through language, verbal training facilitates the expression and refinement of creative concepts (Li et al., 2023). This verbal articulation helps individuals clarify their thinking and fosters collaboration and idea exchange among participants, ultimately enhancing overall creative output (Romdhani et al., 2024). *Thinking training*, which focuses on improving cognitive flexibility, problem-solving abilities, and divergent thinking, follows closely behind in effectiveness (Rute-Pérez et al., 2024). Through exercises such as brainstorming, lateral thinking puzzles, and mind mapping, thinking training stimulates mental agility and encourages participants to explore unconventional solutions to challenges. By challenging cognitive biases and expanding mental horizons, thinking training cultivates a fertile ground for creative insights and breakthroughs (Ramezani & Fawcett, 2024). *Emotional training*, while valuable in fostering empathy, emotional intelligence, and self-awareness, tends to yield comparatively weaker results in creativity enhancement. Regarding emotional training, directly translating emotional insights into tangible creative outputs may be challenging (Thompson et al., 2024). While emotional intelligence is important in the creative process, its impact on generating novel ideas or innovative solutions may be less immediate or apparent compared with verbal and thinking training methods. The similarity in the effectiveness levels of technique-based, task-based, and combined training approaches stems from their complementary nature and holistic approach to creativity development. *Technique-based training* focuses on teaching specific creative techniques or methods, such as visualization techniques or ideation frameworks, which can provide individuals with concrete tools for generating and refining ideas (Schöbel et al., 2024). *Task-based training* immerses participants in real-world creative challenges or projects, allowing them to apply and integrate various creative skills and approaches in practical contexts (Hsu, 2024). *Combined training*, which integrates elements of both technique-based and task-based approaches, offers a comprehensive and balanced

approach to creativity development, leveraging the strengths of each method to maximize creative potential. In summary, the variation in effectiveness among the creativity training approaches is attributable to their respective emphases on verbal expression, cognitive flexibility, emotional intelligence, and the practical application of creative skills. While verbal and thinking training methods effectively foster creative thinking and expression, emotional training may not be as effective in yielding tangible creative outcomes. Meanwhile, technique-based, task-based, and combined training approaches offer diverse yet equally effective pathways to creativity enhancement, catering to different learning preferences and objectives.

5.2. Conclusions

An analysis conducted in this study, comprising 109 effect sizes derived from 29 studies, reveals that creativity can, on average, be enhanced through various forms of training programs. First, we assessed publication bias and heterogeneity in the literature and provided an overall effect size for the study. Our findings indicated a moderate correlation between the training program and creativity among university students, and the training program exhibited a significant impact on creativity ($p < 0.05$). Second, the influence of the training program was most pronounced on creative thinking, with relatively similar effects observed on graphic creativity and verbal creativity. However, the impact on emotional creativity was the weakest. Third, the results of the moderator variable analysis indicated that longer training durations do not necessarily yield better results. Projects exceeding 3 years should focus on enhancing students' latent abilities and shifting their modes of thinking. The training was more effective for graduate students and was most effective when conducted individually. Verbal training exhibited the highest effectiveness in fostering student creativity, followed by thinking training, while emotional training yielded the least favorable outcomes.

6. Implications and limitations

6.1. Implications

6.1.1. Theoretical implications

Firstly, the results demonstrate that verbal training yields the most significant outcomes, underscoring the critical role of cognitive skill development in creativity training. This finding aligns with the conclusions of Gu et al. (2022a) and offers a robust theoretical foundation for the formulation of creativity training programs. Specifically, the design of such programs should prioritize interventions that target cognitive processes—such as divergent thinking and problem-solving skills—thereby facilitating improvements in students' perception, attention, memory, executive function, logical reasoning, and processing speed.

Secondly, the study reveals that combined training exerts a positive influence on students' creativity, highlighting its potential to enhance creativity across multiple dimensions (Zhan et al., 2023). By engaging several components of creativity simultaneously, integrated training approaches may prove more effective than isolated methods. In conclusion, the design of creativity training curricula should adopt a comprehensive approach, taking into account the synergistic effects of diverse training strategies.

Thirdly, bringing a new methodological perspective to the field, we have determined the effect sizes of different types of creativity training, different academic levels (undergraduates and graduates), and different activity types (group and alone) through subgroup analysis. Using meta-regression analysis, we have identified the optimal duration for creativity training, guiding researchers in designing programs and courses by integrating relevant theories. This includes strategies for maintaining students' attention over a long period and providing appropriate stimulation or intervention to achieve better outcomes in creativity training.

6.1.2. Practical implications

Firstly, this study offers valuable insights for educators, prompting a re-evaluation of pedagogical strategies and curriculum design (Wang et al., 2023). Educators should optimize instructional methodologies by prioritizing approaches that actively cultivate cognitive skills. This involves reconfiguring curricular content to move away from traditional didactic teaching, with an increased focus on embedding creativity training into educational frameworks. Furthermore, teachers should be encouraged to explore and implement innovative teaching practices that enhance creativity.

Secondly, the critical role of educators in fostering creativity is underscored. Teacher development should therefore include training on effectively delivering creativity training (Gu et al., 2022b). Educational institutions are urged to provide continuous professional development opportunities to equip teachers with the necessary skills. Given the demonstrated efficacy of creativity training in enhancing students' creative capacities, educational leaders must allocate the requisite resources and institutional support to facilitate this process.

Thirdly, creativity training programmes should be designed to encourage interdisciplinary collaboration, enabling students to forge connections and engage in cooperative learning across diverse fields of study (Weng et al., 2023). Additionally, the integration of emerging technological tools into creativity training should be considered, to drive educational reform and leverage intelligent technologies to enhance students' creativity (Huang & Chang, 2023).

6.2. Limitations

This investigation considered four moderating variables associated with training programs, factors such as cognitive-based training require further investigation. Future research endeavors could explore these facets and their influence on the effectiveness of creativity. Furthermore, while this study focused on the influence of training programs on university students' creativity, future research

could explore comprehensive analyzes from alternative perspectives. Such investigations will facilitate a more comprehensive understanding of the efficacy of training programs in fostering university students' creativity and offer valuable insights for enhancing educational practices. Despite the considerable effort required, researchers should continue to advocate for continued expansion of creativity training research, to obtain more robust and insightful findings.

Note: Asterisked references indicate articles that provided one or more studies for inclusion in the meta-analysis. : Baruah and Paulus 2008; Campion and Levita 2014; Chiu 2015; Dow and Mayer 2004; Fink et al. 2015; Fleury et al. 2020; Gu et al. 2023; Hargrove and Nietfeld 2015; Karakelle 2009; Karwowski and Soszynski 2008; Kienitz et al. 2014; Lewis and Lovatt 2013; Mahasneh and Gazo 2019; Malycha and Maier, 2017; Meinel et al. 2019; Morin et al. 2018; O'Connor et al. 2016; Onarheim and Friis-Olivarius 2013; Oppezzo and Schwartz 2014; Puccio et al. 2020; Ritter and Mostert 2017; Robbins and Kegley 2010; Stolaki and Economides 2018; Vally et al. 2019; Wang and Horng 2002; West et al. 2012; Zampetakis et al. 2008; Zitek and Vincent 2015

Ethical approval

This research project has been approved by the Ethics Review Committee of the School of Education, Guangzhou University, by the principle of ethics (GZHU2024001). Informed consent was taken from all participants included in this study.

Funding

This study was supported by the 2023 Special Task Project of Humanities and Social Sciences Research by the Ministry of Education, China (Grant No. 23JDSZ3200); 2024 Postgraduate Innovation Capacity Development Programme (Basic Innovation Project) by Guangzhou University, China (Grant No. JCCX2024-009).

CRediT authorship contribution statement

Huifen Guo: Writing – review & editing, Writing – original draft. **Zhen Zhou:** Software, Data curation, Conceptualization. **Fengqi Ma:** Supervision.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Acknowledgments

We appreciate Professor Ma's support and financial assistance during the manuscript-writing process. We extend our heartfelt gratitude to Dr. Zhou for their invaluable assistance in this study's data-collection process. Their contributions have been invaluable to our research endeavor.

Appendix A

Table 4
Overview of publications included in the analysis and their characteristics.

Study name	Country	Type of measure	Type of administration	Overall sample size (EG/CG)	Activity for CG ^{a, b, c, d}	Duration	Activity type	Training procedure
Baruah and Paulus (2008)	United States	OSVM	EG/CG	EG = 11/CG = 11	<i>a</i>	<1 week	Group	Combined training
Campion and Levita (2014)	United Kingdom	SEES, TTCT	EG/CG	EG = 56/CG = 56	<i>b</i>	<2 months	Alone	Verbal training
Chiu (2015)	China (Taiwan)	OTT	EG/CG	EG = 39/CG = 39	<i>b</i>	<1 week	Alone	Combined training
Fink et al. (2015)	Austria	AUT	EG/CG	EG = 24/CG = 29	<i>b</i>	<2 months	Group	Verbal training
Hargrove and Nietfeld (2015)	United States	RAT, MAI	Pre/Posttest	EG = 118/CG = 88	<i>b</i>	<4 months	Alone	Thinking training
Karakelle (2009)	Turkey	OSVM	Pre/Posttest	EG = 15/CG = 15	<i>c</i>	<4 months	Alone	Combined training
Kienitz et al. (2014)	United States	TTCT-F, NEO-FFI	EG/CG	EG = 15/CG = 13	<i>b</i>	<2 months	Alone	Verbal training
Lewis and Lovatt (2013)	United Kingdom	AUT	EG/CG	EG = 21/CG = 20	<i>b</i>	<1 week	Group	Verbal training

(continued on next page)

Table 4 (continued)

Study name	Country	Type of measure	Type of administration	Overall sample size (EG/CG)	Activity for CG ^a , _{b, c, d}	Duration	Activity type	Training procedure
Malycha and Maier (2017)	Germany	SEES, TTCT	EG/CG	EG = 80/CG = 80	<i>c</i>	<1 month	Group	Technique-based training
Meinel et al. (2019)	Germany	ATTA	EG/CG	EG = 59/CG = 59	<i>b</i>	<2 months	Group	Thinking training
Morin et al. (2018)	Canada	CEDA	Pre/Posttest	EG = 59/CG = 59	<i>c</i>	>1 year	Alone	Technique-based training
O'Connor et al. (2016)	Australia	WRT	Pre/Posttest	EG = 62/CG = 62	<i>d</i>	<1 week	Alone	Combined training
Study name	Country	Type of measure	Type of administration	Overall sample size (EG/CG)	Activity for CG ^a , _{b, c, d}	Duration	Activity type	Training procedure
Onarheim and Friis-Olivarius (2013)	Denmark, Canada	AUT	EG/CG	EG = 50/CG = 50	<i>b</i>	<2 months	Group	Thinking training
Oppezzo and Schwartz (2014)	United States	AUT	EG/CG	E = 48/C = 48	<i>b</i>	<1 week	Alone	Combined training
Ritter and Mostert (2017)	The Netherlands	AUT, RAT	Pre/Posttest	EG = 31/CG = 31	<i>b</i>	<1 week	Group	Technique-based training
Wang and Horng (2002)	China (Taiwan)	TTCT	EG/CG, Pre/Posttest	EG = 21/CG = 18	<i>b</i>	<1 year	Group	Thinking training
West et al. (2012)	United States	TTCT	EG/CG, Pre/Posttest	EG = 93/CG = 93	<i>b</i>	<1 year	Group	Combined training
Zitek and Vincent (2015)	United States	RAT	EG/CG	EG = 97/CG = 97	<i>b</i>	<1 month	Group	Combined training
Zampetakis et al. (2008)	Greece	OSVM	EG/CG	EG = 52/CG = 42	<i>b</i>	<4 months	Alone	Combined training
Lubart (2014)	France	EPOC	EG/CG	EG = 63/CG = 63	<i>b</i>	<1 week	Group	Combined training
Gu et al. (2023)	China	AUT	EG/CG, Pre/Posttest	EG = 41/CG = 39	<i>b</i>	<4 months	Group	Combined training
Mahasneh and Gazo (2019)	Jordan	ECI	EG/CG	EG = 35/CG = 32	<i>b</i>	<1 week	Alone	Emotional training
Puccio et al. (2020)	Canada	TTCT	EG/CG	EG = 11/CG = 10	<i>b</i>	<3 year	Group	Technique-based training
Fleury et al. (2020)	France	GAU, RAT	EG/CG	EG = 16/CG = 16	<i>c</i>	<1 week	Alone	Technique-based training
Dow and Mayer (2004)	United States	OSVM	EG/CG	EG = 11/CG = 11	<i>b</i>	<1 month	Group	Combined training
Karwowski and Soszynski (2008)	Poland	TCI, TCT-DP	EG/CG, Pre/Posttest	O = 15	<i>b</i>	<2 months	Alone	Combined training
Robbins and Kegley (2010)	United States	TTCT	EG/CG, pre/posttest	O = 51	<i>b</i>	<4 months	Alone	Thinking training
Stolaki and Economides (2018)	Greece	TTCT	EG/CG, Pre/Posttest	O = 81	<i>b</i>	<1 month	Group	Combined training
Vally et al. (2019)	United Arab Emirates	TTCT	EG/CG, Pre/Posttest	O = 133	<i>e</i>	<4 months	Group	Task-based training

Note. MSLQ = Motivated Strategies for Learning Questionnaire; SEES = Subjective Exercise Experiences Scale; TTCT = Torrance Tests of Creative Thinking; OTT = Overinclusive Thinking Training Task; AUT = Alternative Uses Task; RAT = Remote Association Test; MAI = Metacognitive Awareness Inventory; NEO-FFI = NEO Five-Factor Inventory; ATTA = Abbreviated Torrance Test for Adults; CEDA = Creative Engineering Design Assessment; WRT = Welch Reorganization Test; PSI = Figural-A and Problem Solving Inventory; EPOC = Evaluation of the Potential of Creativity; ECI = Emotional Creativity Inventory; GAU = Guilford's Alternate Uses Test; TCI = Test of Creative Imagination; TCT-DP = Test of Creative Thinking and Drawing Production; CAQ = Creative Achievement Questionnaire; OSVM = originality scale from validated measures.

EG = experimental group; CG = control group; O = overall sample size.

a = no training; *b* = received a different type of training; *c* = no training until after posttest; *d* = training with emotional intelligence; *e* = training with neuro creativity.

Data availability

The authors will make the raw data supporting this study's conclusions available (dora777guo@gmail.com) without undue reservation.

References

- Amabile, T. M. (1983). The social psychology of creativity: A componential conceptualization. *Journal of Personality and Social Psychology*, 45(2), 357–376. <https://doi.org/10.1037/0022-3514.45.2.357>
- Baruah, J., & Paulus, P. B. (2008). Effects of training on idea generation in groups. *Small Group Research*, 39(5), 523–541. <https://doi.org/10.1177/1046496408320049>
- Borenstein, M., Cooper, H., Hedges, L., & Valentine, J. (2009). *Effect sizes for continuous data*. In Cooper, H., Hedges, L.V., & Valentine, J.C. The handbook of research synthesis and meta-analysis, 2nd Ed. (pp. 1–615). Russell Sage Foundation., 2, 221–235.
- Borenstein, M., Hedges, L. V., Higgins, J. P., & Rothstein, H. R. (2010). A basic introduction to fixed-effect and random-effects models for meta-analysis. *Research Synthesis Methods*, 1(2), 97–111. <https://doi.org/10.1002/jrsm.12>
- Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2021). *Introduction to meta-analysis*. Wiley. 2nd ed.
- Bruner, J. S. (1962). The conditions of creativity. Eds.. In H. E. Gruber, G. Terrell, & M. Wertheimer (Eds.), *Contemporary approaches to creative thinking: A symposium held at the university of Colorado* (pp. 1–30). Atherton Press. <https://doi.org/10.1037/13117-001>
- Campion, M., & Levita, L. (2014). Enhancing positive affect and divergent thinking abilities: Play some music and dance. *The Journal of Positive Psychology*, 9(2), 137–145. <https://doi.org/10.1080/17439760.2013.848376>
- Card, N. A. (2012). *Applied meta-analysis for social science research*. The Guilford Publications.
- Chahar, B., Hatwal, V., & Sen, S. (2019). Employees training and its impact on learning and creativity: Moderating effect of organisational climate. *Problems and Perspectives in Management*, 17(2), 430–439. [https://doi.org/10.21511/ppm.17\(2\).2019.33](https://doi.org/10.21511/ppm.17(2).2019.33)
- Chen, Y.-S., Chang, C.-H., Yeh, S.-L., & Cheng, H.-I. (2014). Green shared vision and green creativity: The mediation roles of green mindfulness and green self-efficacy. *Quality & Quantity*, 1–16. <https://doi.org/10.1007/s11135-014-0041-8>
- Cheung, H. N., & Hui, A. N. N. (2023). Impact of a mindfulness-based program on creativity and mindfulness disposition of school children. *Psychology of Aesthetics, Creativity, and the Arts*, 17(6), 694–704. <https://doi.org/10.1037/aca0000598>
- Chiu, F. C. (2015). Improving your creative potential without awareness: Overinclusive thinking training. *Thinking Skills and Creativity*, 15, 1–12. <https://doi.org/10.1016/j.tsc.2014.11.001>
- Cohen, J. (1977). *Statistical power analysis for the behavioural sciences*. Lawrence Erlbaum Associates, Inc. <https://doi.org/10.4324/9780203771587>. Rev. ed.
- Cropley, D. H. (2016). Creativity in engineering. Eds.. In G. E. Corazza, & S. Agnoli (Eds.), *Multidisciplinary contributions to the science of creative thinking* (pp. 155–173). Singapore: Springer
- Davis, G. A. (2004). *Creativity is forever* (pp. 39–42). Kendall Hunt Publishing Company, 5th Edition.
- Dodds, A. E., Reid, K. J., Conn, J. J., Elliott, S. L., & McColl, G. J. (2010). Comparing the academic performance of graduate and undergraduate-entry medical students. *Medical Education*, 44(2), 197–204. <https://doi.org/10.1111/j.1365-2923.2009.03559.x>
- Dow, G. T., & Mayer, R. E. (2004). Teaching students to solve insight problems: Evidence for domain specificity in creativity training. *Creativity Research Journal*, 16(4), 389–398. <https://doi.org/10.1080/10400410409534550>
- Dwan, K., Altman, D. G., Arnaiz, J. A., Bloom, J., Chan, A. W., Cronin, E., & Williamson, P. R. (2008). Systematic review of the empirical evidence of study publication bias and outcome reporting bias. *PLoS One*, 3(8), e3081. <https://doi.org/10.1371/journal.pone.0003081>
- Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *BMJ (Clinical Research ed.)*, 315(7109), 629–634. <https://doi.org/10.1136/bmj.315.7109.629>
- Findlay, C. S., & Lumsden, C. J. (1988). The creative mind: Toward an evolutionary theory of discovery and innovation. *Journal of Social and Biological Structures*, 11(1), 3–55. [https://doi.org/10.1016/0140-1750\(88\)90025-5](https://doi.org/10.1016/0140-1750(88)90025-5)
- Fink, A., Benedek, M., Koschutnig, K., Pirker, E., Berger, E., Meister, S., & Weiss, E. M. (2015). Training of verbal creativity modulates brain activity in regions associated with language and memory-related demands. *Human Brain Mapping*, 36(10), 4104–4115. <https://doi.org/10.1002/hbm.22901>
- Finke, R. A., Ward, T. B., & Smith, S. M. (1992). *Creative cognition: Theory, research, and applications*. The MIT Press.
- Fleith, D. D. S., Renzulli, J. S., & Westberg, K. L. (2002). Effects of a creativity training program on divergent thinking abilities and self-concept in monolingual and bilingual classrooms. *Creativity Research Journal*, 14(3–4), 373–386. https://doi.org/10.1207/S15326934CRJ1434_8
- Fleury, S., Agnès, A., Vanukuru, R., Goumillout, E., Delcombel, N., & Richir, S. (2020). Studying the effects of visual movement on creativity. In *Thinking skills and creativity*, 36, Article 100661. <https://doi.org/10.1016/j.tsc.2020.100661>
- Freud, S. (1910). *The Origin and development of psychoanalysis 1910* (Vol. 21). University of Illinois Press. No. 2.
- Freud, S. (1964). *Leonardo da vinci and a memory of his childhood*. WW Norton & Company.
- Grégoire, D. A., Corbett, A. C., & McMullen, J. S. (2011). The cognitive perspective in entrepreneurship: An agenda for future research. *Journal of management studies*, 48(6), 1443–1477. <https://doi.org/10.1111/j.1467-6486.2010.00922.x>
- Gu, X., Ritter, S. M., Delfmann, L. R., & Dijksterhuis, A. (2022b). Stimulating creativity: Examining the effectiveness of four cognitive-based creativity training techniques. *The Journal of Creative Behavior*, 56(3), 312–327. <https://doi.org/10.1002/jobc.531>
- Gu, X., Ritter, S. M., & Dijksterhuis, A. (2024). Online creativity training: Examining the effectiveness of a comprehensive training approach. *International Journal of Technology and Design Education*, 34(1), 403–426. <https://doi.org/10.1007/s10798-023-09820-2>
- Gu, X., Tan, Y., Wu, X., Cai, Z., Lai, Q., Cheng, M., & Zhao, Q. (2022a). Active versus passive strategy in online creativity training: How to best promote creativity of students with different cognitive styles? *Thinking Skills and Creativity*, 44, Article 101021. <https://doi.org/10.1016/j.tsc.2022.101021>
- Gu, X., Tong, D., Shi, P., Zou, Y., Yuan, H., Chen, C., & Zhao, G. (2023). Incorporating STEAM activities into creativity training in higher education. *Thinking Skills and Creativity*, 50, Article 101395. <https://doi.org/10.1016/j.tsc.2023.101395>
- Gutman, L. M., & Schoon, I. (2013). *The impact of non-cognitive skills on outcomes for young people: Literature review*. London, England: Education Endowment Foundation (EEF) and Institute of Education, University of London.
- Hargrove, R. A., & Nietfeld, J. L. (2015). The impact of metacognitive instruction on creative problem-solving. *The Journal of Experimental Education*, 83(3), 291–318. <https://doi.org/10.1080/00220973.2013.876604>
- Hennessey, B. A., & Amabile, T. M. (2010). Creativity. *Annual Review of Psychology*, 61(1), 569–598. <https://doi.org/10.1146/annurev.psych.093008.100416>
- Higgins, J. P., Thompson, S. G., Deeks, J. J., & Altman, D. G. (2003). Measuring inconsistency in meta-analyses. *BMJ (Clinical Research ed.)*, 327(7414), 557–560. <https://doi.org/10.1136/bmj.327.7414.557>
- Hsu, H. C. (2024). Pretask training for web-based second language collaborative writing. *Language Learning & Technology*, 28(1), 1–30. <https://hdl.handle.net/10125/73551>
- Huang, H. T., & Chang, Y. S. (2023). Effects of virtual reality on creative performance and emotions: A study of brainwaves. *Computers in Human Behavior*, 146, Article 107815. <https://doi.org/10.1016/j.chb.2023.107815>
- Im, S., Bhat, S., & Lee, Y. (2015). Consumer perceptions of product creativity, coolness, value and attitude. *Journal of Business Research*, 68(1), 166–172. <https://doi.org/10.1016/j.jbusres.2014.03.014>
- Kansizoglu, H. B. (2017). The effect of graphic organizers on language teaching and learning areas: A meta-analysis study. *Education and Science*, 42(191), 139–164. <https://doi.org/10.15390/EB.2017.677>
- Karakelle, S. (2009). Enhancing fluent and flexible thinking through the creative drama process. *Thinking Skills and Creativity*, 4(2), 124–129. <https://doi.org/10.1016/j.tsc.2009.05.002>
- Karwowski, M., & Soszynski, M. (2008). How to develop creative imagination?: Assumptions, aims and effectiveness of Role Play Training in Creativity (RPTC). *Thinking Skills and Creativity*, 3(2), 163–171. <https://doi.org/10.1016/j.tsc.2008.07.001>
- Khalil, R., Godde, B., & Karim, A. A. (2019). The link between creativity, cognition, and creative drives and underlying neural mechanisms. *Frontiers in Neural Circuits*, 13, 18. <https://doi.org/10.3389/fncir.2019.00018>

- Kienitz, E., Quintin, E. M., Saggat, M., Bott, N. T., Royalty, A., Hong, D. W. C., & Reiss, A. L. (2014). Targeted intervention to increase creative capacity and performance: A randomized controlled pilot study. *Thinking Skills and Creativity*, 13, 57–66. <https://doi.org/10.1016/j.tsc.2014.03.002>
- Kolayis, H., Turan, H., & Ulusoy, Y. O. (2012). Comparison of problem-solving disposition of students in physical education teacher and psychological counselling and guidance. *Procedia-Social and Behavioral Sciences*, 46, 1939–1942. <https://doi.org/10.1016/j.sbspro.2012.05.407>
- Kugley, S., Wade, A., Thomas, J., Mahood, Q., Jørgensen, A. M. K., Hammerstrøm, K., & Sathe, N. (2016). Searching for studies: A guide to information retrieval for Campbell. *Campbell Systematic Reviews*, 10. Retrieved from http://www.campbellcollaboration.org/images/Campbell_Methods_Guides_Information_Retrieval.pdf
- Lewis, C., & Lovatt, P. J. (2013). Breaking away from set patterns of thinking: Improvisation and divergent thinking. *Thinking skills and creativity*, 9, 46–58. <https://doi.org/10.1016/j.tsc.2014.03.002>
- Li, M. H., Tai, C. H., Luh, J. J., Chen, Y. J., Hsu, W. L., & Lee, Y. Y. (2023). Influence of verbal instruction on gait training in Parkinson's disease: A randomized controlled trial. *American Journal of Physical Medicine & Rehabilitation*, 10–1097. <https://doi.org/10.1097/PHM.0000000000002420>
- Liu, M., Pang, W., Guo, J., & Zhang, Y. (2022). A meta-analysis of the effect of multimedia technology on creative performance. *Education and Information Technologies*, 27(6), 8603–8630. <https://doi.org/10.1007/s10639-022-10981-1>
- Liu, Z., Yu, H., Feng, M., & Hou, Y. (2023). Thinking styles and creativity: The mediating role of psychological adjustment in college students. *Behavioral Sciences*, 13(10), 875. <https://doi.org/10.3390/bs13100875>
- López-Nicolás, R., López-López, J. A., Rubio-Aparicio, M., & Sánchez-Meca, J. (2022). A meta-review of transparency and reproducibility-related reporting practices in published meta-analyses on clinical psychological interventions (2000–2020). *Behavior Research Methods*, 54(1), 334–349. <https://doi.org/10.3758/s13428-021-01644-z>
- Lucchiari, C., Sala, P. M., & Vanutelli, M. E. (2019). The effects of a cognitive pathway to promote class creative thinking. An experimental study on Italian primary school students. *Thinking Skills and Creativity*, 31, 156–166. <https://doi.org/10.1016/j.tsc.2018.12.002>
- Ma, H. (2006). A synthetic analysis of the effectiveness of single components and packages in creativity training programs. *Creativity Research Journal*, 18, 435–446. https://doi.org/10.1207/s15326934crj1804_3
- Mahasneh, A. M., & Gazo, A. M. (2019). Effect of the training program to improve emotional creativity among undergraduate students. *Psychology in Russia: State of the Art*, 12(3), 137–148. <https://doi.org/10.11621/pir.2019.0310>
- Malycha, C. P., & Maier, G. W. (2017). The random-map technique: Enhancing mind-mapping with a conceptual combination technique to foster creative potential. *Creativity Research Journal*, 29(2), 114–124. <https://doi.org/10.1080/10400419.2017.1302763>
- Mansfield, R. S., Busse, T. V., & Kreplka, E. J. (1978). The effectiveness of creativity training. *Review of Educational Research*, 48, 517–536. <https://doi.org/10.3102/00346543048004517>
- Matraeva, A. D., Rybakova, M. V., Vinichenko, M. V., Oseev, A. A., & Ljapunova, N. V. (2020). Development of creativity of students in higher educational institutions: Assessment of students and experts. *Universal Journal of Educational Research*, 8(1), 8–16. <https://doi.org/10.13189/ujer.2020.080102>
- Mayer, R. E. (2006). The role of domain knowledge in creative problem solving. Eds. In J. C. Kaufman, & J. Baer (Eds.), *Creativity and reason in cognitive development* (pp. 150–152). Cambridge University Press. <https://doi.org/10.1017/CBO9780511606915.010>
- Meinel, M., Wagner, T. F., Baccarella, C. V., & Voigt, K. I. (2019). Exploring the effects of creativity training on creative performance and creative self-efficacy: Evidence from a longitudinal study. *The Journal of Creative Behavior*, 53(4), 546–558. <https://doi.org/10.1002/jocb.234>
- Moher, D., Shamseer, L., Clarke, M., Ghersi, D., Liberati, A., Petticrew, M., & Prisma-P Group. (2015). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews*, 4, 1–9. <https://doi.org/10.1186/2046-4053-4-1>
- Moreau, D., & Gamble, B. (2022). Conducting a meta-analysis in the age of open science: Tools, tips, and practical recommendations. *Psychological Methods*, 27(3), 426–432. <https://doi.org/10.1037/met0000351>
- Morin, S., Robert, J. M., & Gabora, L. (2018). How to train future engineers to be more creative? An educative experience. *Thinking Skills and Creativity*, 28, 150–166. <https://doi.org/10.1016/j.tsc.2018.05.003>
- Murad, M. H., Chu, H., Lin, L., & Wang, Z. (2018). The effect of publication bias magnitude and direction on the certainty in evidence. *BMJ Evidence-Based Medicine*. <https://doi.org/10.1136/bmjebm-2018-110891>
- Nandi, P. L., Chan, J. N. F., Chan, C. P. K., Chan, P., & Chan, L. P. K. (2000). Undergraduate medical education: Comparison of problem-based learning and conventional teaching. *Hong Kong Medical Journal*, 6(3), 301.
- O'Connor, P. J., Gardiner, E., & Watson, C. (2016). Learning to relax versus learning to ideate: Relaxation-focused creativity training benefits introverts more than extroverts. *Thinking Skills and Creativity*, 21, 97–108. <https://doi.org/10.1016/j.tsc.2016.05.008>
- Onarheim, B., & Friis-Olivarius, M. (2013). Applying the neuroscience of creativity to creativity training. *Frontiers in Human Neuroscience*, 7, 656. <https://doi.org/10.3389/fnhum.2013.00656>
- Oppizzo, M., & Schwartz, D. L. (2014). Give your ideas some legs: The positive effect of walking on creative thinking. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 40(4), 1142. <https://doi.org/10.1037/a0036577>
- Osborn, A. (1963). *Applied imagination*. New York: Scribner's.
- Park, S. J. (2023). Testing the effects of a TRIZ invention instruction program on creativity beliefs, creativity, and invention teaching self-efficacy. *Education and Information Technologies*, 28(10), 12883–12902. <https://doi.org/10.1007/s10639-023-11614-x>
- Parnes, S. J. (1967). *Creative behaviour guidebook*. New York: Scribner's.
- Paulus, P. (2000). Groups, teams, and creativity: The creative potential of idea-generating groups. *Applied Psychology*, 49(2), 237–262. <https://doi.org/10.1111/1464-0597.00013>
- Pigott, T. D., & Polanin, J. R. (2020). Methodological guidance paper: High-quality meta-analysis in a systematic review. *Review of Educational Research*, 90(1), 24–46. <https://doi.org/10.3102/0034654319877153>
- Plucker, J. A., Beghetto, R. A., & Dow, G. T. (2004). Why isn't creativity more important to educational psychologists? Potentials, pitfalls, and future directions in creativity research. *Educational Psychologist*, 39(2), 83–96. https://doi.org/10.1207/s15326985Sep3902_1
- Puccio, G. J., Burnett, C., Acar, S., Yudes, J. A., Holinger, M., & Cabra, J. F. (2020). Creative problem-solving in small groups: The effects of creativity training on idea generation, solution creativity, and leadership effectiveness. *The Journal of Creative Behavior*, 54(2), 453–471. <https://doi.org/10.1002/jocb.381>
- Puccio, G. J., Mance, M., & Murdock, M. C. (2010). *Creative leadership: Skills that drive change*. Sage Publications.
- Ramezani, M., & Fawcett, A. J. (2024). Cognitive-motor training improves reading-related executive functions: A randomized clinical trial study in dyslexia. *Brain Sciences*, 14(2), 127. <https://doi.org/10.3390/brainsci14020127>
- Rampersad, G., & Patel, F. (2014). Creativity as a desirable graduate attribute: Implications for curriculum design and employability. *Asia-Pacific Journal of Cooperative Education*, 15(1), 1–11. Available <https://eric.ed.gov/?id=EJ1113642>
- Ritter, S. M., Gu, X., Crijns, M., & Biekens, P. (2020). Fostering students' creative thinking skills by means of a one-year creativity training program. *PloS One*, 15(3), Article e0229773. <https://doi.org/10.1371/journal.pone.0229773>
- Ritter, S. M., & Mostert, N. (2017). Enhancement of creative thinking skills using a cognitive-based creativity training. *Journal of Cognitive Enhancement*, 1, 243–253. <https://doi.org/10.1007/s41465-016-0002-3>
- Robbins, T. L., & Kegley, K. (2010). Playing with Thinkertoys to build creative abilities through online instruction. *Thinking Skills and Creativity*, 5(1), 40–48. <https://doi.org/10.1016/j.tsc.2009.07.001>
- Romdhani, Z., Ceylan, H. I., Hammami, R., Sahli, F., Dhahbi, W., Souissi, N., & Nikolaidis, P. T. (2024). Small-sided games with coaches' verbal encouragement have a positive effect on aerobic performance, mood state, satisfaction and subjective effort in male semi-professional soccer players. *Journal of Men's Health*, 20(1), 1–10. <https://doi.org/10.22514/jomh.2024.018>
- Rose, L. H., & Lin, H.-T. (1984). A meta-analysis of long-term creativity training programs. *Journal of Creative Behavior*, 18, 11–22.
- Rosenthal, R. (1979). The file drawer problem and tolerance for null results. *Psychological Bulletin*, 86(3), 638–641. <https://doi.org/10.1037/0033-2909.86.3.638>
- Ruiz-del-Pino, B., Fernandez-Martin, F. D., & Arco-Tirado, J. L. (2022). Creativity training programs in primary education: A systematic review and meta-analysis. *Thinking Skills and Creativity*, 46, Article 101172. <https://doi.org/10.1016/j.tsc.2022.101172>

- Rute-Pérez, S., Rodríguez-Domínguez, C., Sánchez-Lara, E. M., Pérez-García, M., & Caracuel, A. (2024). Does including Facebook training improve the effectiveness of computerized cognitive training? A randomized controlled trial. *Clinical Rehabilitation*, Article 02692155241228832. <https://doi.org/10.1177/02692155241228832>
- Said-Metwally, S., Fernández-Castilla, B., Kyndt, E., Van den Noortgate, W., & Barbot, B. (2021). Does the fourth-grade slump in creativity actually exist? A meta-analysis of the development of divergent thinking in school-age children and adolescents. *Educational Psychology Review*, 33(1), 275–298. <https://doi.org/10.1007/s10648-020-09547-9>
- Samstad, G. I. (1962). General Electric's creative courses. Eds., In S. J. Parnes, & H. F. Harding (Eds.), *A source book for creative thinking*. New York: Charles Scribner's Sons
- Schöbel, T., Schuschke, L., Youssef, Y., Rotzoll, D., Theopold, J., & Osterhoff, G. (2024). Immersive virtual reality in orthopaedic surgery as elective subject for medical students: First experiences in curricular teaching. *Die Orthopädie*, 1–9. <https://doi.org/10.1007/s00132-024-04491-w>
- Scott, G., Leritz, L. E., & Mumford, M. D. (2004). The effectiveness of creativity training: A quantitative review. *Creativity Research Journal*, 16, 361–388. <https://doi.org/10.1080/10400410409534549>
- Şendag, S., Yakin, İ., & Gedik, N. (2023). Fostering creative thinking skills through computer programming: Explicit or integrated teaching? *Education and Information Technologies*, 28(9), 10819–10838. <https://doi.org/10.1007/s10639-023-11629-4>
- Sio, U. N., & Lortie-Forgues, H. (2024). The impact of creativity training on creative performance: A meta-analytic review and critical evaluation of 5 decades of creativity training studies. *Psychological bulletin*. <https://doi.org/10.1037/bul0000432>
- Sternberg, R. J. (1986). *Intelligence applied: Understanding and increasing your intellectual skills*. San Diego, CA: Harcourt Brace Jovanovich.
- Stolaki, A., & Economides, A. A. (2018). The Creativity Challenge Game: An educational intervention for creativity enhancement with the integration of information and Communication technologies (ICTs). *Computers & Education*, 123, 195–211. <https://doi.org/10.1016/j.compedu.2018.05.009>
- Thompson, C. M., Voorhees, H. L., Taniguchi-Dorios, E., Makos, S., Pool, K., & Babu, S. (2024). Development and initial assessment of an emotional support provision training intervention for interpersonal support providers in the context of chronic illness. *Health Communication*, 1–14. <https://doi.org/10.1080/10410236.2024.2325183>
- Torrance, E. P. (1972). Can we teach children to think creatively? *Journal of Creative Behavior*, 6, 114–143. Available <https://files.eric.ed.gov/fulltext/ED061544.pdf>.
- Valgeirsdottir, D., & Onarheim, B. (2017). Studying creativity training programs: A methodological analysis. *Creativity and Innovation Management*, 26(4), 430–439. <https://doi.org/10.1111/caim.12245>
- Vally, Z., Salloum, L., AlQedra, D., El Shazly, S., Albloshi, M., Alsheraifi, S., & Alkaabi, A. (2019). Examining the effects of creativity training on creative production, creative self-efficacy, and neuro-executive functioning. *Thinking Skills and Creativity*, 31, 70–78. <https://doi.org/10.1016/j.tsc.2018.11.003>
- Van der Zanden, P. J., Meijer, P. C., & Beghetto, R. A. (2020). A review study about creativity in adolescence: Where is the social context? *Thinking Skills and Creativity*, 38, Article 100702. <https://doi.org/10.1016/j.tsc.2020.100702>
- Von Fange, E.K. (1963). Professional creativity (Y. Kato & K. Okamura, (Trans.)). Tokyo: Iwanami Shoten.
- Wallas, G. (1926). The art of thought. J. Cape: London.
- Wang, C. W., & Horng, R. Y. (2002). The effects of creative problem-solving training on creativity, cognitive type and R&D performance. *R&D Management*, 32(1), 35–45. <https://doi.org/10.1111/1467-9310.00237>
- Wang, S., Sun, Z., & Chen, Y. (2023). Effects of higher education institutes' artificial intelligence capability on students' self-efficacy, creativity and learning performance. *Education and Information Technologies*, 28(5), 4919–4939. <https://doi.org/10.1007/s10639-022-11338-4>
- Weisberg, R. W. (1993). *Creativity: Beyond the myth genius*. New York: Freeman.
- Weng, X., Ng, O. L., Cui, Z., & Leung, S. (2023). Creativity development with problem-based digital making and block-based programming for science, technology, engineering, arts, and mathematics learning in middle school contexts. *Journal of Educational Computing Research*, 61(2), 304–328. <https://doi.org/10.1177/07356331221115661>
- West, R. E., Tateishi, I., Wright, G. A., & Fonoimoana, M. (2012). Innovation 101: Promoting undergraduate innovation through a two-day boot camp. *Creativity Research Journal*, 24(2–3), 243–251. <https://doi.org/10.1080/10400419.2012.677364>
- Xing, Z., & Qi, Y. (2023). Development of creativity in physical education teachers using interactive technologies: Involvement and collaboration. *Education and information technologies*, 28(5), 5763–5777. <https://doi.org/10.1007/s10639-022-11422-9>
- Xu, F. (2013). *New creative education: When creative thinking, entrepreneurial education, and innovative education come together*. In Shavinina, L.V. (Ed.) The routledge international handbook of innovation education, (pp. 172–180). Routledge.
- Yeh, Y. C., & Lin, C. F. (2015). Aptitude-treatment interactions during creativity training in e-learning: How meaning-making, self-regulation, and knowledge management influence creativity. *Journal of Educational Technology & Society*, 18(1), 119–131. <https://www.jstor.org/stable/jeductechsoci.18.1.119>
- Zampetakis, L. A., Moustakis, V., Dewett, T., & Zampetakis, K. (2008). A longitudinal analysis of student creativity scripts. *The Journal of Creative Behavior*, 42(4), 237–254. <https://doi.org/10.1002/j.2162-6057.2008.tb01298.x>
- Zhan, Z., Yao, X., & Li, T. (2023). Effects of association interventions on students' creative thinking, aptitude, empathy, and design scheme in a STEAM course: Considering remote and close association. *International Journal of Technology and Design Education*, 33(5), 1773–1795. <https://doi.org/10.1007/s10798-022-09801-x>
- Zhang, C., Zhou, Y., Wijaya, T. T., Chen, J., & Ning, Y. (2024). Effects of a problem posing instructional interventions on student learning outcomes: A three-level meta-analysis. *Thinking Skills and Creativity*, 53, Article 101587. <https://doi.org/10.1016/j.tsc.2024.101587>
- Zitek, E. M., & Vincent, L. C. (2015). Deserve and diverge: Feeling entitled makes people more creative. *Journal of Experimental Social Psychology*, 56, 242–248. <https://doi.org/10.1016/j.jesp.2014.10.006>